Use case UC3: Reschedule or Cancel Appointment
Primary Actor: Authorized clinic staff
Stakeholder and Interests:

Patient:

- Wants to ensure their confidentiality.
- Wants to keep referral, if applies.
- Wants their appointment cancelled or reschedule to a date that suits them.

Authorized clinic staff/Clinic:

- Want to easily cancel an appointment.
- Want to easily reschedule an appointment while ensuring no schedule conflicts or infringed applicable scheduling constraints.
- Want to automatic notifications to all parties concerned upon appointment change.
- Want to keep logs of all appointment changes.

Healthcare professional:

- Wants to have a conflict-free schedule.

Preconditions:

- User (patient or employee with appointment managing rights) is authenticated.
- The patient has an existing record in the database.
- The appointment exists in the database.
- The appointment is up and coming, not a past one or an already cancelled one.
- In case of reschedule, the new time slot requested is free in the healthcare professional's schedule.

Success Guarantee (Postconditions):

- Canceled or rescheduled appointment's state in the database is updated.
- Canceled or rescheduled appointment doesn't appear in the schedule.
- If rescheduled, new appointment is recorded in the database.
- If rescheduled, new appointment appears in the schedule.

Main success scenario (or basic flow):

1. The patient contacts an authorized clinic staff (user) to cancel or reschedule an appointment, and the authorized clinic staff opens the schedule.
2. The user finds the appointment and opens it.
3. The user clicks on cancel or reschedules.
4. The system changes the state of the appointment.
5. The system frees that time slot on the schedule.

It's done for cancelations. The following steps are for a rescheduling scenario:

6. The system displays the appointment details and the available time slots of the healthcare professional.
7. The user informs the patient of the available time slots, and the patient chooses a new one.
8. The user selects the new time slot.
9. The system validates the applicable scheduling constraints.
10. The system displays a summary of the new appointment and asks for confirmation.

11. The user confirms.
12. The system records the new appointment and fills the slot in the schedule.

Failure scenario (alternative flow):

6a. The healthcare professional has no available time slots. The system notifies the user and offers the option to cancel the appointment instead.

7a. The user selects a new time slot that violates a scheduling constraint (triggered in step 9.) The system notifies the user of the reason and doesn't let them continue until a valid time slot is selected.

Special Requirements:

- Admin staff see appointment data only, no clinical information.
- System flags should appear using formatted red text in the terminal

Technology and Data Variations List:

- User credentials, schedule, appointment list.
- Date and time entered in UTC.

Open Issues:

- Is there a minimum notice period to modify an appointment?
- Can the patient reschedule with a different professional instead?
- Notification system for updates on an appointment.
- Should a patient be able to cancel or reschedule themselves?